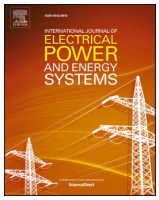
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Enabling mass integration of electric vehicles through distributed energy resource management systems

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A B S T R A C T

Global initiatives are actively progressing to integrate large numbers of electric vehicles (EVs) as part of efforts to electrify and decarbonize the transportation sector. This approach plays a crucial role in addressing global greenhouse emissions and pollution problems. However, mass integration of EVs imposes unprecedented challenges to electrical power systems, especially to their distribution grids. These challenges range from planning issues, where the traditional panning procedures are not adequate for integration of EVs and their charging stations, to operational challenges, where a huge number of dispersed EVs can cause peak load increase, voltage violations, congestions, feeder overloads, etc. Hence, despite the significant importance of maximizing the deployment of EVs, their stochastic integration can give rise to critical challenges in electric power systems, thereby impeding their subsequent expansion. Therefore, in order to properly integrate mass amounts of EVs, without causing technical violations in the electrical power system, intelligent software tools are required, to aid distribution system operators and engineers in the transition to a new era of electrified transportation. In this paper, we argue that novel software solutions called Distributed Energy Resource Management Systems (DERMSs) are a key solution for enabling a safe integration of mass amounts of EVs into emerging distribution grids. Specifically, we introduce a novel Hybrid DERMS framework, as an integration of multiple hierarchical levels of DER management solutions, and propose this solution as a suitable instrument for distribution system operators and grid engineers to safely integrate and manage high volumes of EVs within the grid infrastructure. The hypothesis is validated through a real-life use case for integration and management of high amounts of EVs into an existing distribution grid, using an industrial DERMS solution. Specifically, an in-depth analysis of adding two large charging stations, boasting capacities of 5 MW each, has been performed through all stages of integration, from the planning stage, up to using their flexible output to manage technical constraint violations in real time. The influence of these EV charging stations on both the feeder where they are planned to be integrated, as well as on the entire supply substation, have been analyzed in detail. The resultant findings, especially on successfully managing overloads, peak load increases, as well as voltage violations through intelligent use of aggregated EVs through DERMS, have proven to be notably encouraging, further substantiating the paper's hypothesis.

1. Introduction

Electric vehicles (EVs) are widely considered as the future of transportation and supplanting combustion engine vehicles rapidly across the globe. In the last decade, EV technology rapidly evolved, with huge improvements in charging time, battery capacity, range, etc., overcoming the main barriers that kept drivers from purchasing EVs [1–4]. Moreover, the price of EVs, even though still higher than traditional vehicles in many parts of the world, has considerably dropped in the last decade [5]. Considering that the traditional combustion engine vehicles contribute to about one fifth of air pollution in the world, and that the EV technology is ever evolving towards solving the aforementioned barriers for entry, there are numerous initiatives from the government bodies around the globe to deploy as much EVs as possible and to gradually completely electrify and decarbonize the transportation sector [1–5]. However, the main advantages for EV drivers (fast charging, larger batteries, etc.), create new challenges for security and reliability of electrical power systems, and especially their distribution grids [6–8].

Traditional distribution grids were designed and built with a premise of being passive networks, with a sole aim of distributing electrical energy from the supply substation to end customers [9,10]. Thus, from the inception of modern power systems more than a century ago, until the last decade, distribution grids were managed and operated as passive networks with one-way power flow (from the supply substation to end customers). In such an environment, distribution network operators (DNOs) and planning engineers successfully managed their grids using

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simple, not so automated processes [10]. However, in the last two decade, an increase in deployment of distributed energy resources (DERs) and behind the meter resources, is rapidly transforming traditional distribution networks into complex and dynamically changing systems, especially because of EVs and the temporal and spatial uncertainty they introduce [10]. Management of these emerging power systems using traditional tools and processes is highly unlikely to succeed. This gives rise to critical looming challenges in electric power systems, thereby impeding their subsequent expansion.

Fast EV chargers, in ranges of hundreds of kilowatts, if deployed under optimized and randomly – which seems to be the case right now, may cause serious issues to power system's security and reliability [11]; peak load would unquestionably rise in parts with high penetration of EV chargers, voltage profiles may fluctuate due to unmanaged charging of EVs, legacy feeders designed with a premise of annual load rise in a range of several percent may become insufficient and frequently overloaded due to fast charging of EVs, etc. [11]. Moreover, traditional planning procedures designed for passive distribution grids are inapplicable for planning mass integration of EVs and their accompanying charging infrastructure. Namely, due to an extremely dynamic nature of EV charging, in order to properly plan their integration and to successfully design and place their charging equipment, without causing violations of technical constraints, electrical power system's planning tools should be tightly integrated with tools for real-time management of these dynamically changing grids [11].

A review of the state-of-the-art on integration of EVs into electrical power systems suggests the existence of a decent body of work that addresses some of these challenges. Hedegaard et al. analyzes the effects of integrating various shares of EVs into power system of Northern European countries [12]. The authors simulate the addition of EVs in a range from 2.5 % to 53 % of total vehicles and detect several challenges to power systems' operation when the penetration of EVs becomes significant. The main challenges they identify are peak load rise, voltage fluctuations, and feeder overloads. However, they conclude that if EV charging is intelligently planned and managed, EVs could be successfully used as flexibility resources for distribution system operator's (DSO's) applications. Yang et al. provide a comprehensive review of computational scheduling methods for integrating plug-in EVs into power systems [13]. The authors analyze scheduling methods, formulated as optimization problems containing various objectives and constraints. They conclude that even though traditional planning methods are inapplicable for EVs, there are promising solutions in the available literature. However, most of these methods are not well verified and need further testing on real systems. Mahmud et al. investigates integration of EVs and their management through internet of energy (IoE) [14]. They review EV technologies, their connectivity, impacts on grid and standards required for the efficient and economic operation of EVs with DERs in the IoE, as well as connectivity issues of mass amounts of EV chargers. The authors conclude that an intelligent software tool is required to successfully integrate and manage emerging distribution grids with mass amounts of EVs along with other DERs. Zhang et al. propose a methodology for optimizing power system's demand by intelligently using EV charging [15]. They conclude that this methodology provides better results when fast charging is implemented. However, they identify other challenges due to fast charging, such as peak load rise, voltage fluctuations, power losses, harmonics, etc. Thus, a comprehensive tool that would consider all the challenges imposed by EVs is required. Falvo et al. analyze integration of EVs into demand response programs [16]. They present benefits of using EVs as demand response resources, but do not investigate possible challenges in the process, such as availability of EVs, capacity of feeders, etc. Clairand et al. provide a review of challenges and potential solutions for integrating high penetration of electric busses and electric taxis into power systems [17]. The authors identify peak load rise as the biggest challenge and present several pilot projects that are currently under way and investigate different methodologies for safely integrating mass amounts of EVs. Nour et al. provide a comprehensive review of positive and negative impacts of EV charging on electric power systems [11]. They identify that uncontrolled charging of EVs may cause increase of peak demand, voltage deviation from the acceptable limits, phase unbalance due to single-phase chargers, harmonics distortion, overloading of the power system equipment, and increase of power losses. On the other hand, the authors claim that if proper tools were to be developed, to intelligently control and optimize EV charging, these challenges could be transformed into numerous benefits, such as frequency and voltage regulation, reactive power compensation, and congestion management. Casella et al. investigate optimization models for the integration of EVs and charging stations in the power system [18]. The authors review optimization methods for both planning procedures for EV integration, as well as for operational management. They conclude that in order to efficiently integrate and safely manage large fleets of EVs, digitization of control centers with proper software tools are needed for planning and management of electrical power systems, that would be able to jointly manage EVs, production systems, storage, and loads, taking into account new technologies (such as vehicle-to-grid (V2G)) and the constraints of the electrical grid. Pecas Lopes et al. propose a conceptual framework for integration of EVs into power systems, considering technical operation of the grid and electricity markets [19]. The authors state that a centralized EV charging control strategy is required for the integration of large amounts of EVs. Further, if the EV charging is properly coordinated, in a centralized way, charging/discharging schedules can be optimized to provide various benefits to grid operators, such as voltage improvements, congestion management, etc. However, a centralized software solution is required to do so without violating technical constraints in the grid. Khosrojerdi et al. provide a review of technical challenges to power system operation due to integration of EVs [20]. Similar to the previous studies, they conclude that the biggest challenges are peak load rise, voltage fluctuations, voltage imbalance, and harmonics. On the other hand, if properly managed, in a centralized fashion, intelligent charging could provide operational benefits and valuable flexibility resources to DSOs. Finally, Tan et al. analyze V2G technology, review state-of-the-art in optimization techniques for V2G, and provide their view on how it can benefit the electrical grid operation [21]. They conclude that V2G is a promising technology that could provide numerous benefits to DSOs, but that it needs to be properly managed and coordinated by all stakeholders, such as power utilities, V2G aggregators and EV owners. From the abovementioned literature, it is evident that uncontrolled and stochastic integration of EVs and their accompanying charging infrastructure would be harmful to the grid and could destabilize electrical distribution grids. On the other hand, preliminary pilot projects and new studies show that if this integration is properly planned, and consequently intelligently managed in a centralized fashion and in conjunction with other DERs such as solar PVs, wind turbines and energy storages, EVs can actually present valuable flexibility resources for DSOs. This proposition serves as the motive for this research.

This paper lays out the importance of novel comprehensive software tools for planning and management of dynamically changing distribution grids, called Distributed Energy Resource Management Systems (DERMS), that could successfully be implemented in practice. As the name implies, in most simple terms, a DERMS is a software solution for management of emerging distribution systems with high penetration of DERs. In this context, DERs are generator agnostic, allowing the management of distributed generators of all kinds (solar PVs, wind turbines, diesel gensets, etc.), energy storages, and suitable for advanced functionalities such as demand response and energy efficiency programs, but also EVs and EV charging stations [22–26]. There exist various DER management solutions, each with completely unique architectures and aimed to accomplish different goals with using DERs, that are all frequently called DERMS. This paper focuses on how each of these DER management solutions can aid in proper planning, integration, and management of high amounts of EVs and their charging infrastructure,

into emerging electrical distribution systems. Moreover, this paper demonstrates the integration of different hierarchical levels of DER management solutions, into a so-called "Hybrid DERMS", as a unique tool for DSOs to properly plan and consequently manage mass amounts of EVs and their charging infrastructure.

The motivation for this paper is a clear need for comprehensive research on how to intelligently plan and manage high amounts of EVs in emerging distribution grids, and what are the available tools that will aid DSOs to actively manage their distribution grids with increasing amounts of EVs, in a coordination with other DERs. The contributions of this paper are threefold. First, it systematically identifies and comprehensively discusses the technical barriers that impede the seamless integration of a large number of EVs into power systems. It provides a rigorous analysis of these challenges, shedding light on critical issues that must be addressed to enable effective EV integration. Second, a Hybrid DERMS is introduced as a tool to overcome these issues and substantially contribute to allowing integration and active management of high amounts of EVs. Per knowledge of the authors, this is the first paper that introduces a Hybrid DERMS as a tool for enabling EV integration and management, and that computationally substantiates these claims. Third, it highlights that widespread adoption of the concepts presented herein has the capacity to lead to the decarbonization of the transportation sector. This, in turn, holds the promise of generating substantial reductions in pollution levels and greenhouse gas emissions, thereby addressing pressing environmental and sustainability challenges. The hypothesis put forth in this paper is validated and its effectiveness is evaluated by implementation in an industrial-grade DERMS solution. Specifically, it is employed to plan the integration of two large-scale EV charging stations, boasting capacities of 5 MW each, into an existing distribution feeder. Subsequently, the utilization of these charging stations as flexible resources, actively contributing to the management and optimization of the feeder's operations are demonstrated.

The findings of this paper contribute to a better and more thorough understanding how to combat the main challenges blocking mass EV integration, and consequent decarbonization of the transportation and energy sector. The paper explores the synergistic potential of EVs as mobile energy storage units within DERMS, enabling effective load balancing and grid stabilization. By intelligently managing the charging and discharging of EV batteries based on real-time grid conditions and renewable energy availability, these systems can significantly reduce reliance on fossil-fueled plants, leading to a substantial decrease in greenhouse gas emissions and air pollutants. This integration also supports the transition to a cleaner energy mix, as it encourages the use of renewable energy sources for both EV charging and grid supply. Furthermore, the adoption of DERMS for EV integration fosters resilience in power systems against the adverse impacts of extreme weather events. As extreme weather events become more unpredictable, DERMS can enhance grid reliability by incorporating localized power generation and storage. In this way, they can mitigate power outages during natural disasters, ensuring critical infrastructure and essential services remain operational. Ultimately, the exploration of EV integration through DERMS offers a comprehensive strategy for pollution reduction in power and energy systems, contributing significantly to the sustainable and resilient future of the energy infrastructure. At a high level, the findings of this study could inform the system operators and policy-makers on initiatives related to integration of EVs.

The rest of the paper is organized as follows: Section II lays out planning and operational challenges to distribution system operators and engineers, due to integration of high amounts of EVs; Section III introduces DERMS concept, various types of DERMS solutions available today, and elaborate on strengths and weaknesses of each of them; Section IV dives into the capabilities of DERMS to aid distribution system operators and engineers to integrate and manage mass amounts of EVs, without causing technical violations in distribution grids. In this section also, a Hybrid DERMS is proposed as a proper solution for safe integration and management of distribution grids with high shares of EVs; Section V presents real-life case studies of integration and real-time management of a distribution feeder with a considerable share of EVs, using industrial-grade DERMS solution; Section VI concludes and closes the paper.

2. Challenges to power systems due to mass EV integration

Poorly managed and not well controlled charging of EVs would inevitably cause severe impacts on electrical distribution grids, proportional to the penetration level of EVs [11,27,28]. The term "significant" is a relative quantification, as increasing number of studies show that even 20–30 % of EVs of a total number of vehicles, if charged in an unmanaged way, would cause issues to the grid operation [11,12,29]. In addition, improper planning and random integration of EV charging infrastructure would be very challenging for the future of distribution grids. The main challenges identified in the available literature, caused by stochastic integration and uncontrolled charging of EVs are: (i) increase in the peak load demand, (ii) overloading of transformers, lines, and cables (which considerably shortens their life expectancy), (iii) increase in voltage violations, (iv) increase in system unbalance due to single phase chargers, and (v) a significant increase in power losses because of higher rate of consumption [11,27,28]. Moreover, if traditional planning procedures were to be followed for planning the integration of mass amounts of EVs, huge infrastructural investments would be required, which would further defer their mass rollout.

This section discusses planning and operational challenges, respectively, imposed to electrical distribution grids by integration of EVs and their charging infrastructure.

2.1. Planning challenges

Traditional distribution grid planning was a linear and a straightforward process [30,31]. Planning departments within distribution utilities would examine the system conditions, apply traditional load forecasting to predict the future load growth (and some very limited impact of an insignificant DER penetration, at that time), and finally run analysis on the system needs, reinforcements, and replacements [30,31]. Utilities would further determine the best solution, such as upgrading substations and/or lines and cables, and draft an investment plan. As the load growth was limited to several percent annually, this process was sufficiently accurate, investments were fairly limited, and it worked well for decades. Fig. 1 summarizes the process, in accordance with [31].

However, the case has radically changed with the increasing amounts of EVs and their accompanying charging infrastructure. The latest EV chargers (so-called level 3 chargers), mostly used for commercial charging stations, each are up to 400 kW of power [11,29]. This means that a charging station comprised of several level 3 chargers is in a range of several megawatts of power, which is a considerable load for distribution network [11,29]. Given the capacity required to meet this demand, planning the addition of multiple charging stations to a distribution substation's area or a feeder is not an easy task. If traditional procedures were to be followed, adding several charging stations of this magnitude (which is absolutely required for even partially electrified transportation [11,29]), would require huge investments in changing or reinforcing almost all of the grid elements.

For example, California's new state regulation requires 35 % of new car models sold in 2026 to be zero-emission, ramping up to 100 % in 2035 [29]. Powering these vehicles (with keeping lights on for other customers) requires the state to triple its current electric power production, and consequently to properly strengthen all of the affected grid elements [29]. Obviously, huge investments in such projects are not the sole problem, as even if the utility accepts such an investment plan, the entire process of obtaining an array of permits and consequently performing such a laborious project would require years to be done, if traditional planning procedures were to be followed. As a consequence,



Fig. 1. Traditional distribution grid planning [31].

utility companies would be forced to stall further enrollment of EVs until the grid is reinforced to be able to host them without causing technical issues. In other words, rollout of EVs would be deferred for years, if not decades, and electrification and decarbonization of the transportation sector would be radically jeopardized.

This indicates the need for new and advanced planning tools that would enable a way to integrate mass amounts of EVs faster, but still without causing issues to the electrical distribution grid. To do so, active network management (ANM) and non-wires alternatives (NWAs) are being proposed, which offer alternative ways of planning and utilization of all the grid resources, with the flexibility of aggregated DERs and EVs, in order to defer capital investments and enable faster integration of DERs and EVs [22–26]. Later in this paper, Section IV discusses the importance of DERMS software for providing a comprehensive set of ANM and NWA tools aimed to offer DSOs and planning departments a way to intelligently plan and integrate mass amounts of EVs without an immediate need for capital expenditures and accompanying lengthy procedures.

2.2. Operational challenges

As briefly discussed at the beginning of this section, challenges that high amounts of EVs introduce to real time operation of electrical distribution grids can be summarized as in Fig. 2 [11]. This subsection elaborates the impact of uncontrolled charging of EVs on each of these issues.

2.2.1. Peak load increase

As shown in [11,32–35] even low penetration levels of EVs (10 %-30 %), if charged in an unmanaged way, could increase the peak load demand up to almost 60 %. Obviously, with peak load demand being already one of the biggest issues for distribution utilities [29], this large increase could not be accommodated at a reasonable planning horizon and cost. Wang et al. show that even 10 % of uncontrolled EVs would contribute to a considerable peak load rise [32]. A direct consequence of such an increase in peak load demand would be exceeding the grid equipment limits, as shown in [35]. This means that without having advanced solutions for managing the EV charging, and utilizing the flexibility of EVs for DSO's benefits, huge infrastructural investments would be required before even a low penetration of EVs would be feasible to be integrated. On the other hand, with controlled charging and a centralized management of DERs that allows aggregated EV flexibility, faster integration of higher amounts of EVs would be feasible [36,37]. Later in this paper, Section IV introduces a hybrid DERMS solution that can offer DSOs a proper set of tools to intelligently manage EVs and their charging infrastructure, and allow faster EV rollout.

2.2.2. Congestions and overloads

Due to considerable higher load caused by EV charging, much more energy flows through distribution grid infrastructure. This affects all elements, from substations, feeders, MV/LV transformers, and all the other equipment such as cables, breakers, switches, etc. Several studies [38–40] show that even as low as 10 %-50 % of EVs, and without level 3 chargers (only level 1 and level 2 chargers were considered), if charged stochastically, would cause overloading on as much as 50 % of transformers in a distribution grid. Considering that level 3 chargers consume more than 10 times the power utilized by level 2 chargers, it is reasonable to assume that with level 3 chargers the situation with overloading would be even more severe. Considerable amounts of level 3 charging stations, with unmanaged usage of chargers, would cause congestion inside the distribution grid, overloading of almost all the elements on the power flow paths, and would destabilize the grid. Thus, only if intelligently planned, and properly utilized in orchestration with other flexible resources such as controllable DERs, would it be feasible to add considerable amounts of EV charging stations, without overloading and congesting the distribution grid and its elements.

2.2.3. Voltage regulation

Voltage regulation is one of the most important power quality measurements in electrical distribution grids; voltage magnitudes need to be maintained within a standard range at all times, that is plus/minus 5 % (in some special cases up to 10 %) of the rated values. Even without EVs, voltage regulation is a challenging task [38,41,42,54]. With EVs as huge electricity consumers, keeping voltages inside boundaries, especially on long feeders, would not be easy. Studies have shown that even less than 30 % of EVs could cause voltage issues on a secondary grid [38,42]. Moreover, with single-phase chargers installed at customers' houses, voltage imbalance will become also an issue to consider. UI-Haq, Shahnia et al. show that even if only level 1 and level 2 chargers are considered, and less than 50 % of EV penetration, significant voltage imbalance would occur in distribution grids [43,44]. Finally, with V-2-G technology, where EVs would supply the grid at times when they are not

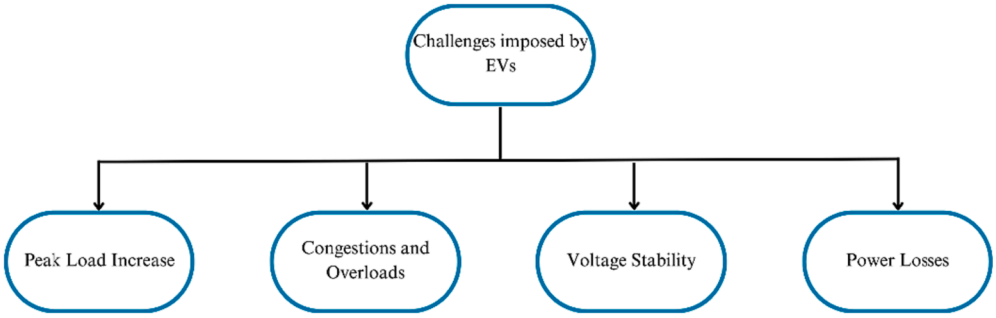


Fig. 2. Operational challenges due to mass integration of EVs.

used and when the grid needs extra power, a new phenomenon of voltage rises above allowed limits can occur in grid nodes where the chargers are connected. Thus, a comprehensive grid management solution is required to successfully cope with these new challenges and keep voltages inside boundaries. Section IV will discuss how a hybrid DERMS solution is able to deliver such capability.

2.2.4. Power losses

Similar to the above, significantly higher load, caused by EV charging, causes much higher currents to flow through transformers, lines, and cables, causing power losses to increase drastically. Numerous studies show that even less than 50 % of EV penetration would cause unacceptable power losses, assuming unmanaged charging [45–47]. Fernandes et al. show that as low as 30 % of EV penetration could cause such large losses that huge investments in reinforcing grid elements would be needed [45]. Pillai et al. show on an example of a Danish power network that a penetration of 50 % of EVs, if charged in an uncontrolled way, could cause increase in power losses for up to 40 % [46]. Masoum et al. studied the impact that 40 % of EVs would have on transformer losses, and found out that losses could go up to 300 % [47]. Also, it is important to note at this place that in all the mentioned studies, only level 1 and level 2 chargers were investigated, so it reasonable to consider when the penetration of level 3 chargers becomes significant, the situation with power losses could only get worse. However, all of the referenced studies agree that managed charging, through an intelligent software solution, would be able to keep power losses in order [11].

Next, the DERMS concept is revisited, where different types of DERMS solutions are discussed, and a novel Hybrid DERMS concept is proposed as an integration of different hierarchical levels of DER management solutions.

3. DER management software solutions and the proposed hybrid DERMS framework

The DERMS, as the name implies, is a software solution for managing DERs, such as solar PVs, wind turbines, EVs, EV charging stations, energy storages, etc. [22–26]. However, it is still an emerging concept, and simply saying "DERMS" may not be sufficient, as there are multiple different DER management solutions, aimed to achieve different goals for different stakeholders. This issue has been discussed in depth in [22–24], and it is summarized in Table 1. In this table, first, the key stakeholders interested in deploying DER management software are listed. Second, the goals that each of these entities desires to achieve with DER management software are briefly explained. Third, the hierarchical structure of the software itself, required to achieve the desired goal, is marked besides each of the listed solutions. Finally, in the last column of the table, each solution is titled differently, therefore distinguishing these different solutions per their types and characteristics [22].

In the following, the roles and responsibilities of each of these software solutions within the context of DERMS are explained.

Table 1
Different Types of DER Management Solutions.

Stakeholder	Goals aimed to be achieved	Architecture	Title
Distribution system operator, Planning and innovation departments in distribution utilities	Relieve congestions and voltage violations, solve reverse power flow problems, defer network reinforcement, enhance distribution grid hosting capability, optimization of existing and new assets, secure grid edge stability.	Centralized; aware of the accurate network model	Utility/Grid DERMS
Prosumers, microgrid participants, individual EVs, behind-the-meter DERs	Aggregate EVs and DERs for: local energy management, including microgrids, to optimize costs (energy charges), integrate DGs, and improve resiliency	Decentralized and local; not aware of the accurate network model	DER Aggregator, EV Aggregator, Microgrid Operator
Market participants, EV aggregators, DER aggregators	Wholesale: optimize portfolio (risk management) and minimize imbalance costs Retail: hedge against high wholesale prices, support new retail tariffs, and competitive differentiation	Decentralized; not aware of the accurate network model	Local Electricity Market Operator

3.1. Utility DERMS

A Utility DERMS (or sometimes called "Grid DERMS") is a centralized software solution, aimed to be used by distribution grid operators, grid engineers, and network planners, to aid them in planning, control, protection, and overall management of emerging distribution grids with high penetration of DERs and EVs [22–26]. It is a grid-aware solution, based on a highly accurate network model and real-time state estimation, for providing awareness of as-operated grid conditions at all times [22–26]. In most cases, a Utility DERMS consists of planning, real-time, and lookahead modules. These modules are consisted of sophisticated functionalities to enable planning of new DER and EV integrations, what-if scenario analysis, and NWAs, on one hand, and real-time and look-ahead applications for control, optimization, and protection of dynamically-changing distribution grids, on the other. Thus, a Utility DERMS solution provides a complete set of tools to aid utilities in an entire process, from planning the integration of a new DER or an EV charging station, to incorporating these new devices into the distribution grid's eco-system, and using them as flexible resources for various control and optimization needs. A conceptual high-level depiction of a Utility DERMS' architecture is depicted in Fig. 3 [22].

As can be seen in Fig. 3, in addition to modeling traditional grid elements, a Utility DERMS needs to accurately model all DERs and the entire EV charging infrastructure (with their dynamical behavior), and with an appropriate representation that corresponds to the timescale of interest [22]. On top of an accurate network model, to provide awareness of accurate grid conditions in real time, a Utility DERMS must contain sophisticated power flow and state estimation algorithms that are constantly supplied by real-time Supervisory Control and Data Acquisition (SCADA) and Advanced Metering Infrastructure (AMI) measurements, load and generation forecasts, EV schedules, real-time topology, and as-operated switching state from the field, as well as planned schedules from EVs and DERs based on their market participation. Further, on top of the core applications for network analysis, a Utility DERMS shall deploy advanced applications for control, optimization, and protection of distribution grids with high amount of DERs and EV chargers, dispersed all over the grid. Thus, a Utility DERMS is a comprehensive software solution, enabling DSOs much needed tools for proper planning of new DER and EV integrations, complete awareness of the changing distribution grid conditions in real-time, as well as a powerful set of tools to keep dynamically changing grid conditions inside technical boundaries. Finally, as it is aimed for a control room operation, Utility DERMS directly communicates with front-of-the-meter DERs and EV charging stations (mainly connected to the MV

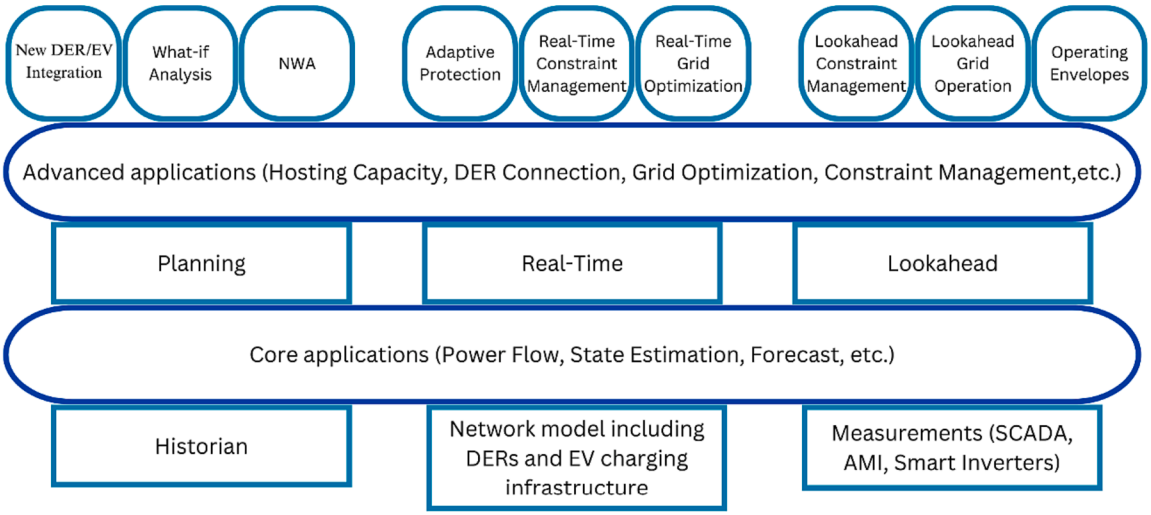


Fig. 3. A Utility DERMS.

grid, for example level 3 charging stations), whereas it reaches behind-the-meter DERs and EVs (most of the level 1 and level 2 chargers) through its communication with DER/EV aggregators, making it aware and able to use the flexibility of all the available resources.

3.2. DER/EV aggregators

DER/EV Aggregators are decentralized software solutions, mainly concerned with aggregating behind-the-meter DERs and EV chargers (level 1 and 2 chargers, in most cases) into groups of large amounts of these small-scale resources, that can then offer a considerable amount of aggregated power and flexibility to a DSO. Moreover, DER/EV Aggregators are responsible for direct communication with individual DERs and EV chargers, and for incentivizing them to participate in smart and controlled energy efficiency and charging programs [11,22]. This means that by providing various monetary and other benefits to DER and EV owners in return for services or access to their assets, DER/EV Aggregators are able to use the flexibility of aggregated DERs and EV chargers in some critical parts of a day, to help DSOs to keep the grid stable. For example, some of the provisions that DER/EV Aggregators offer, include enabling aggregated DER and EV power to participate in the electricity market (and consequently, to be used for balancing, frequency regulation or similar services), provision of demand response and load shifting programs, peak load reduction, and other, mostly customer-related services, that can be used and coordinated for the benefits of the entire power system [11,22]. It is important to note that in order for an EV Aggregator to have access and authority over management of a particular EV or an EV charger, proper flexible contracts between the EV owner and the aggregator need to be in place which become a legal matter hence not within the scope of this paper. Nevertheless, these novel types of contracts, are starting to be particularly popular as they provide highly accelerated way to integrate new EV chargers (or other types of DERs) because the potential constraint violations would be managed by using the flexibility of these resources, instead of waiting for the grid to be strengthen by building new assets [22,23].

The DER/EV Aggregators generally do not have access to an accurate network model and are thus unaware of the grid level technical constraints, such as transformer and lines' overloads, relay settings and coordination, voltage limits, or power losses constraints [22]. Because of this reason, even though services provided by DER/EV Aggregators are highly important as they increase the overall flexibility of distribution grids by enabling the usage of the flexibility of small-scale DERs and level 1 and level 2 EV chargers, in most cases a DER/EV Aggregator cannot be a sole DER management solution, as the schedules and dispatch of DERs and EVs provided by DER/EV Aggregators alone may endanger grid assets, if they are not previously validated by the grid-aware enterprise solution, such as a Utility DERMS [22].

3.3. Microgrid controllers

Nowadays, an increasing number of military objects, hospitals, university campuses, as well as communities with EVs, rooftop solar panels, and household batteries are forming self-sustainable microgrids [48]. The US Department of Energy defines a microgrid as "A group of interconnected loads and DERs with clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid and can connect and disconnect from the grid to enable it to operate in both grid-connected and islanded modes" [48]. Thus, from a DSO's point of view, a Microgrid is a decentralized entity, and a Microgrid Controller is responsible for the proper operation of each microgrid [22]. Consequently, like DER/EV Aggregators, Microgrid Controllers aggregate DERs, EVs, and controllable loads, but unlike DER/EV Aggregators, a microgrid has clearly defined geographical and electrical boundaries. An additional difference is that while DER/EV Aggregators in most cases only offer their services to DSOs, Microgrid Controllers also need to ensure that the microgrid itself is properly operated. However, in the grid-connected mode, multiple networked microgrids, with the excess power of their DERs and stored power of multiple EVs, can offer considerable flexibility to DSOs to be used for their needs—for the benefits of the distribution grid as a whole [22]. These services, in such conditions, are the same as in the case of DER/EV Aggregators. Through proper integration of a grid-aware Utility DERMS, and multiple Microgrid Controllers that are responsible for individual microgrids, DSOs may have enough flexibility to perform NWAs for optimizing the grid, increase the reliability of the grid, as well as defer or even avoid costly and time-consuming network reinforcement [22].

3.4. Local electricity market operators

In the last several years, an increasing number of DER/EV Aggregators have been proposed to harness DER and EV potential to provide market-based grid services that may lower the life-cycle cost of integrating expensive DER and EV technologies [22,49–52]. To enable the participation of DERs and aggregated EVs in the electricity markets, Local Electricity Market Operators play a crucial role [22]. Such entities could help create new opportunities for aggregated DERs and EVs, as well as other small-scale resources (e.g. prosumers) to participate in the market, promoting local energy self-sufficiency, grid resilience, and

integration of EVs and renewable energy resources. However, similar to DER Aggregators, Local Electricity Market Operators are decentralized entities which do not possess an accurate network model and are thus unaware of the impact that the services offered by DER/EV Aggregators would have on the distribution grid assets. To avoid causing damage on distribution grid elements, there should be established communication infrastructure between a grid-aware Utility DERMS and Local Electricity Market Operators, where all the schedules proposed by a market operator would first need to be validated by a Utility DERMS [22].

Summarizing the above, in the next subsection a novel concept of "Hybrid DERMS" will be discussed, as an integration of all hierarchical levels of DER management solutions listed in this section.

3.5. The hybrid DERMS framework

As discussed in previous subsections, all of the listed DER management solutions provide multiple benefits to DSOs and end-customers, but neither of them is able to solve all the challenges that high amounts of EVs and DERs impose. Thus, in order to be able to accurately plan the integration of these dynamically changing resources, forecast their impact on a distribution grid, defer capital investments into building new grid assets, use DERs and EV chargers as flexible resources for distribution grid needs, harness their aggregated potential and monetize their flexibility, this paper proposes an integrated solution that encompasses Utility DERMS, DER/EV Aggregators, Microgrid Controllers, and Local Electricity Market Operators into a so-called "Hybrid DERMS" framework. This integration, specially oriented towards integration and management of high amounts of EVs, is presented in Fig. 4.

Integrated together, DER/EV Aggregators and Microgrid Controllers would significantly enhance DSO's awareness of small-scale, highly dispersed DERs and EVs, and through a Utility DERMS, operators and grid engineers would have a complete awareness of real-time conditions in the grid including the impact that behind-the-meter DERs and EV chargers have on the grid. Moreover, DSOs would be able to utilize the flexibility of all the available resources, including direct communication with large-scale DERs and level 3 chargers, as well as small-scale resources through aggregators, in order to use it to optimize the grid and avoid any technical constraint violations. Additionally, once the regulation allows aggregated DERs and stored energy of multiple EV chargers, to enter electricity markets, integration of a Utility DERMS with DER Aggregators and Local Electricity Market Operators, will be of a high importance, as the schedules of DERs and DER Aggregators, aimed for market participation, would need to be checked and validated against technical grid constraints, which only a Utility DERMS would be able to do. Probably the most challenging aspect for distribution grid management – dynamic behavior of EVs combined with the intermittent nature of renewable DERs, could also be successfully managed through a Hybrid DERMS solution. Using sophisticated load and generation forecast engine, and through communication with SCADA and AMI measurements as well as with DER/EV Aggregators that should always be able to provide as-operated state of their resources, Utility DERMS would be able to implement all the available data and use it to determine real-time conditions as well as predicted state of a distribution grid. Further, for most critical cases of sudden and considerable variability, that cannot be predicted by the Forecast engine, Utility DERMS can predefine interdependent behavior of its variable DERs and EV chargers, and set them to change their output and/or schedules on any variability above the threshold. For instance, consider the scenario where a Utility DERMS is configured to manage the deployment of nearby batteries or stored power from aggregated EVs. If the solar PV generation exceeds a specified threshold, Utility DERMS can initiate the discharge of these resources with a predetermined amount. Additionally, utilizing State Estimation and Constraint Management advanced applications, Utility DERMS would identify fluctuations in the grid and recalibrate the grid conditions, asses the availability of flexible resources, account for technical limitations, and anticipate potential violations. It will then automatically redistribute all available resources to restore the grid to an optimal state free from violations.

The primary conceptual framework presented in this paper revolves around the innovative utilization of Hybrid DERMS as a central theme. The core premise of this research is to leverage the Hybrid DERMS concept to facilitate the seamless integration and dynamic control of

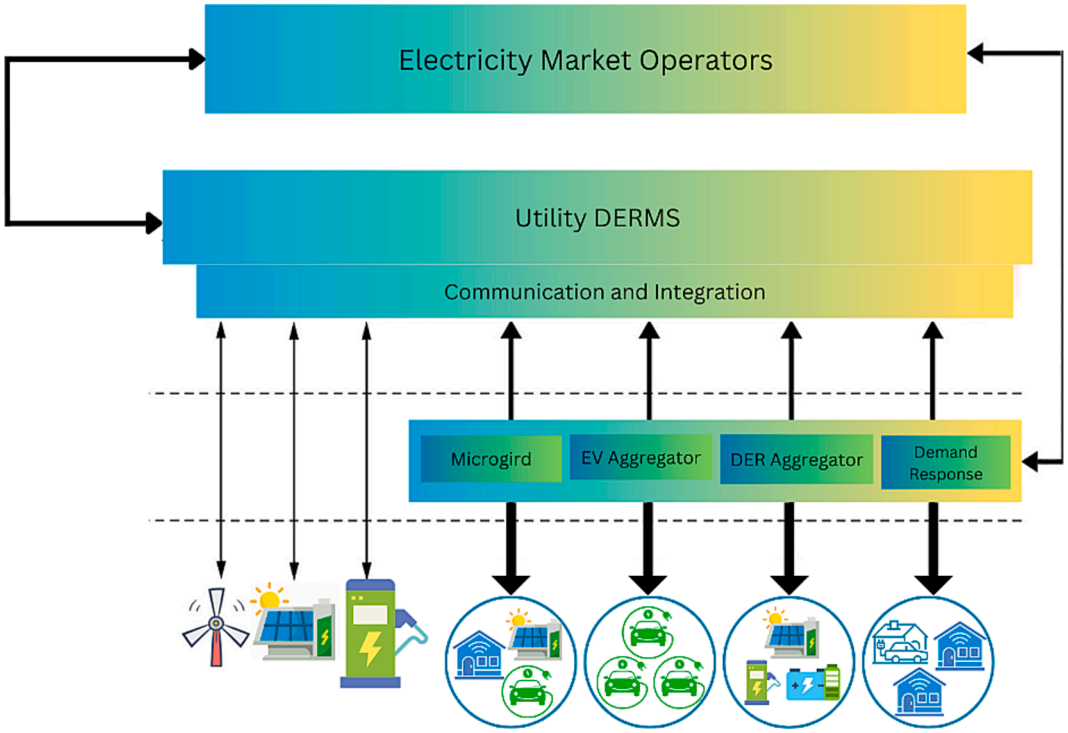


Fig. 4. A Hybrid DERMS Framework.

substantial quantities of EVs within emerging power systems. The ensuing sections of this paper delve deeply into the intricacies of how the implementation of a Hybrid DERMS solution can substantially enhance the strategic aspects of planning and the operational aspects of managing large-scale EV integration. These discussions are further reinforced by the presentation of numerical examples, offering empirical validation and tangible insights into the feasibility and effectiveness of this new approach.

4. Planning and management of high amount of EVs using the proposed hybrid DERMS framework

The primary focus of this section centers on expounding upon the central thesis of this paper. Specifically, it addresses the overarching concept that a Hybrid DERMS constitutes a vital and comprehensive tool essential for orchestrating two key facets: first, the strategic planning associated with integrating EVs and their requisite charging infrastructure into the power ecosystem; second, the adept control and administration of evolving distribution grids grappling with substantial EV penetration, all while maintaining strict adherence to technical compliance standards. In essence, the argument posited in this section firmly establishes that the deployment of a Hybrid DERMS solution emerges as the linchpin capable of enabling the seamless integration and dynamic governance of extensive EV fleets within the burgeoning landscape of emerging power systems.

4.1. Planning the integration of high amounts of EVs through DERMS

The integration of EVs and EV charging infrastructure brings about significant changes in distribution grids, necessitating thorough planning and technical considerations. This integration affects various aspects, including technical and operational processes, regulatory frameworks, and business procedures. To successfully integrate a high penetration of EVs, specialized tools and procedures are required [11–21]. Therefore, a planning module within a Utility DERMS solution becomes essential, providing grid engineers with comprehensive tools to conduct extensive studies on the addition of new EVs and EV charging infrastructure to the existing grid.

A critical aspect of the Utility DERMS planning module is the evaluation of hosting capacity, specifically the ability of the grid to accommodate new EVs and associated charging infrastructure. Hosting Capacity Heat Maps (HCHMs) play a crucial role in this assessment [22]. HCHMs are geographical representations of the grid, where each location is color-coded to indicate its capability to host new EVs. These maps are constructed within the planning module by performing load flow studies on a precise network model. Incremental additions of EV power into network nodes are simulated using a wide range of probabilistic scenarios. The aforementioned load flow studies, with varying charging/discharging scenarios, assess the potential technical violations caused by the introduction of EVs and aim to determine the maximum capacity that the grid can support. HCHMs can be created by considering critical cases, such as peak demand scenarios, or by simulating dynamically changing environments in distribution grids. Once generated, HCHMs are made accessible through a DSO's web portal, providing potential customers and EV developers with valuable information for decision-making. HCHMs serve as the foundation for further comprehensive studies on EV integration.

Upon receiving a specific EV connection request from a customer, EV developer, or when planning a new EV charging facility, in-depth investigations are required. This analysis occurs within the planning module of the Utility DERMS, utilizing advanced planning applications specifically designed for this purpose. The specific Point of Common Coupling (PCC) for the new EV charging station is thoroughly examined, considering parameters such as the required rated power, technology type, charging profiles, and other relevant factors. The impact of the new EV charging station, under different charging scenarios and load profiles, on the rest of the grid, including voltage profiles, power flow patterns, and potential constraints, is carefully evaluated. Accurate network models encompassing distribution transformers, power cables, voltage regulation devices, and other network components are essential for these analyses. Advanced applications, such as optimal power flow algorithms, contingency analysis tools, and stochastic modeling techniques, are employed to simulate and quantify the potential impacts. This comprehensive evaluation enables the identification of potential technical violations and guides decision-making. In cases where the initial analysis indicates the integration of new EVs could lead to grid violations, the planning department must explore alternative options beyond traditional grid asset reinforcement. While traditional reinforcement, involving the resizing of cables and other assets, is a possibility, but as previously discussed, it can be costly and time-consuming. A Utility DERMS should provide a suite of tools to analyze NWAs as viable options for enabling new EV connections. NWAs include innovative strategies that leverage flexible contracts, energy storage (including EVs themselves), and demand response capabilities to mitigate the need for costly network reinforcements. For example, customers can be offered flexible contracts that allow the DSO to curtail excess power during critical periods, or to shift charging schedules to less critical periods, thereby reducing the strain on the grid. Additionally, the feasibility of utilizing energy storage systems, including the battery capacity of connected EVs, and exploring available reverse demand response capabilities to absorb excess power during peak periods should be thoroughly assessed. A Utility DERMS must encompass the necessary tools and algorithms to enable such studies, facilitating the evaluation of NWAs as viable alternatives to traditional grid reinforcement. Finally, after conducting specific EV integration studies and exhausting all available options for increasing hosting capacity, the planning module of the Utility DERMS should enable a multifaceted comparison of various alternatives. This comparison considers policy-related objectives, economic factors, technical feasibility, and reliability requirements over multiple years. Through a rigorous multi-value assessment, the planning department can make informed decisions regarding the integration of new EVs into the grid. The comparison process involves considering factors such as investment costs, operational costs, environmental impacts, grid performance metrics, and customer satisfaction. This comprehensive evaluation aids in selecting the most suitable strategies and solutions for accommodating the increasing penetration of EVs while ensuring grid stability, reliability, and efficiency.

It is evident that the integration of EVs and EV charging infrastructure into distribution grids demands meticulous planning and technical evaluations. The Utility DERMS planning module plays a critical role in assessing hosting capacity, conducting detailed studies, exploring NWAs, and facilitating multi-objective comparisons. By utilizing advanced tools, accurate network models, and sophisticated algorithms, distribution system operators can effectively manage the integration of EVs, optimizing grid operations and reaping the benefits of sustainable transportation technologies. Fig. 5 summarizes the process, in accordance with [22].

4.2. Managing emerging distribution grids with high amounts of EVs through DERMS

After a comprehensive planning process and careful analysis of available options for integrating EVs with the associated charging infrastructure, DSOs must be prepared to actively manage this new infrastructure in conjunction with other resources, such as renewable DERs and energy storage systems. In this operational phase, a Hybrid DERMS plays a crucial role. The operational mode of a Hybrid DERMS enables real-time control, optimization, and protection of the grid through the utilization of the capabilities explained as follows.

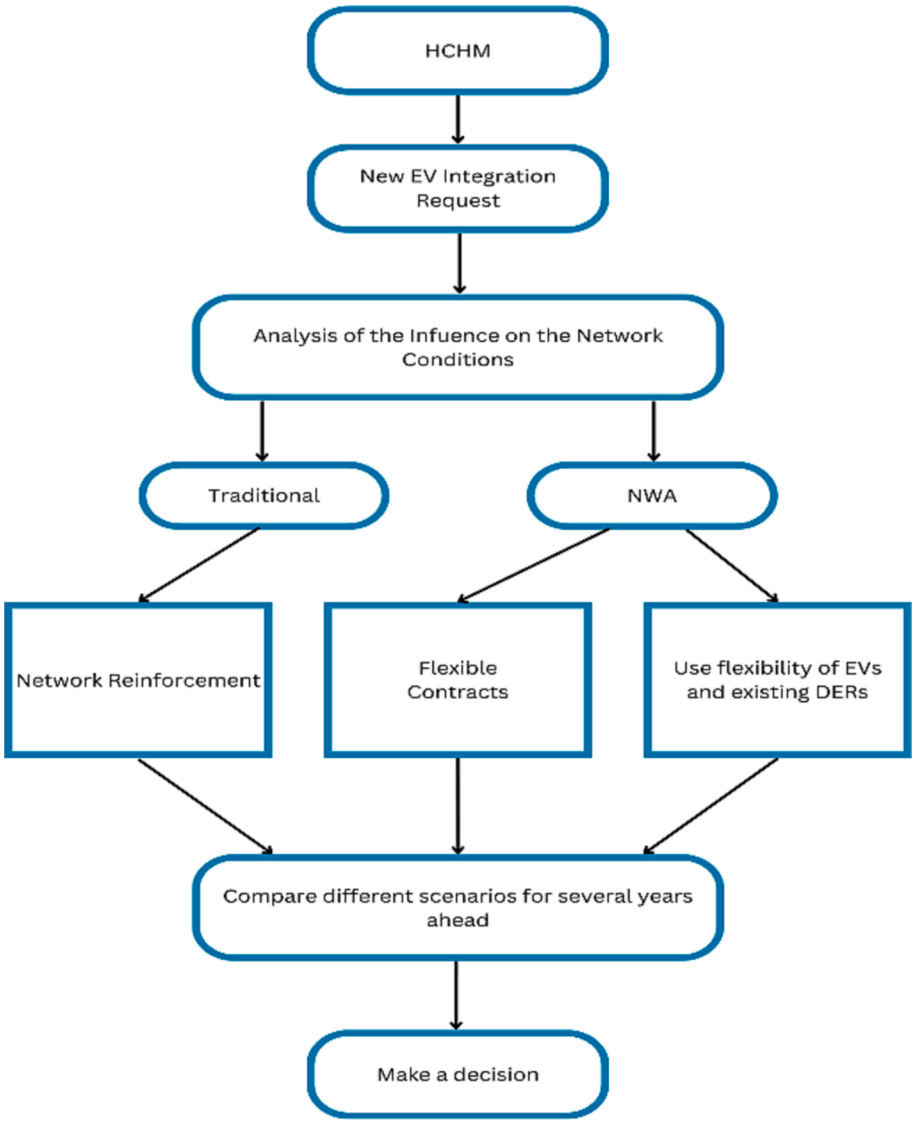


Fig. 5. Planning new EV integration through DERMS.

4.2.1. Real-time monitoring and control

The Hybrid DERMS employs sophisticated techniques for real-time monitoring and control, enabling DSOs to have a detailed understanding of the grid's current conditions. This capability involves continuous data collection, analysis, and decision-making based on the information gathered from various sources, including SCADA systems, Advanced Metering Infrastructure (AMI), and EV charging infrastructure. To ensure effective real-time monitoring, Hybrid DERMS utilizes robust communication networks and protocols to gather data from diverse grid components. SCADA systems provide crucial information such as voltage levels, line currents, and transformer statuses. The AMI, consisting of smart meters installed at customer premises, offers detailed load profiles and consumption patterns (including the influence of level 1 and level 2 chargers on these profiles). The integration with EV charging infrastructure allows the system to access real-time data on EV charging statuses, power levels, and available capacity. The collected data is processed and analyzed in real-time by the Hybrid DERMS. This involves sophisticated algorithms and analytics that handle vast amounts of data, including load and generation forecasts, EV schedules, real-time topology information, and switching state data. By continuously monitoring and analyzing this information, the Hybrid DERMS gains comprehensive situational awareness of the grid's current conditions. With this wealth of real-time data, the Hybrid DERMS can make informed decisions and take proactive control actions to ensure grid security and optimize grid operations. For example, if the system detects an overloaded feeder due to high EV charging demand, it can initiate load-shedding measures or adjust the charging profiles of EVs to alleviate the congestion. Conversely, during periods of low demand and excess generation, the system can encourage EV owners to charge their vehicles, utilizing the available renewable energy resources effectively. Furthermore, the real-time data collected by the Hybrid DERMS facilitates grid forecasting and prediction models. By analyzing historical data alongside real-time information, the system can generate accurate load and generation forecasts, EV charging profiles, and other key metrics. These forecasts play a vital role in optimizing grid operations, as they provide valuable insights for planning and decision-making, allowing DSOs to anticipate grid conditions and proactively manage potential challenges.

In summary, the real-time monitoring and control capabilities of the Hybrid DERMS enable DSOs to gain comprehensive awareness of the electrical grid's current conditions. By continuously collecting and analyzing data from various sources, including SCADA systems, AMI, and EV charging infrastructure, the system empowers operators to make informed decisions and take proactive control actions to optimize grid operations, ensure grid security, and effectively manage the integration of EVs.

4.2.2. Congestion management

One of the key challenges that arise with the integration of a large number of EVs into the distribution grid is the potential for grid congestion. Grid congestion occurs when the electricity demand from EV charging exceeds the available capacity of the grid infrastructure, leading to localized overloads, voltage fluctuations, and compromised system performance. The Hybrid DERMS plays a vital role in addressing this challenge by actively monitoring and managing the congestion levels within the distribution grid. The Hybrid DERMS continuously monitors the grid and analyzes real-time data from various sources, including SCADA systems, AMI, and EV charging infrastructure. By collecting information such as load forecasts, EV charging schedules, and real-time grid conditions, the system gains comprehensive awareness of the current state of the distribution grid. Using this information, the Hybrid DERMS identifies areas within the grid that are at risk of congestion due to high EV charging demand. It analyzes the load profiles, power flow patterns, and capacity constraints of the grid infrastructure to determine potential congestion points. By assessing the availability of resources and capacity limits, the system can identify areas where the grid infrastructure may be strained, and congestion is likely to occur. To mitigate the impact of congestion, the Hybrid DERMS has the capability to implement control strategies and coordination mechanisms. Intelligent charging scheduling is one approach employed, where the system optimizes the charging profiles of EVs to distribute the load more evenly throughout the grid. By considering factors such as EV battery levels, charging requirements, and grid conditions, the system can intelligently schedule charging sessions to avoid simultaneous high-demand periods and reduce localized congestion. Load shifting is another technique used by the Hybrid DERMS to alleviate congestion. It involves incentivizing EV owners to adjust their charging patterns to off-peak or low-demand periods, where the grid has more available capacity. By encouraging EV charging during times when the grid is less stressed, the system can effectively balance the load and reduce the likelihood of congestion. Furthermore, the Hybrid DERMS enables coordination between EV charging and other DERs within the grid. By integrating and orchestrating the charging schedules of EVs with other DERs such as renewable energy sources and energy storage systems, the system can optimize the utilization of available resources and alleviate congestion. For example, during periods of high EV charging demand, the system can prioritize the use of local renewable energy generation or utilize energy stored in batteries to meet the increased load, thereby reducing the strain on the grid infrastructure. The Hybrid DERMS acts as a central control platform that communicates with the EV charging infrastructure, DERs, and other grid components. It exchanges data and signals to coordinate the charging activities and ensure the optimal utilization of resources. The system dynamically adjusts charging rates, initiates load curtailment measures if necessary, and provides real-time feedback to EV owners and operators to manage and balance the charging load in a way that minimizes congestion and ensures reliable grid operation. By actively monitoring and managing congestion levels, implementing intelligent charging scheduling, load shifting, and coordinating EV charging with other DERs, the Hybrid DERMS effectively addresses the challenge of grid congestion associated with a large number of EVs. It enables DSOs to maintain grid reliable power delivery, optimize resource utilization, and ensure safe and efficient grid operations in the presence of high EV penetration.

4.2.3. Grid optimization and control

Grid Optimization and Control is a critical aspect of the Hybrid DERMS, which leverages grid state estimation and real-time data to optimize the dispatch of resources and ensure efficient operation of the distribution grid. By considering the dynamic behavior of EV charging and intermittent renewable DERs, the Hybrid DERMS aims to balance supply and demand while adhering to technical constraints and maintaining grid stability. The Hybrid DERMS utilizes advanced optimization algorithms and control strategies to effectively manage the distribution grid. It takes into account various factors such as grid conditions, load profiles, available generation capacity, and the status of EV charging infrastructure. By analyzing real-time data from SCADA systems, AMI, and EV charging infrastructure, the system gains a comprehensive understanding of the grid's current state. Using this information, the Hybrid DERMS optimizes the dispatch of available resources to address congestion issues and maintain grid stability. It considers the capabilities and characteristics of different resources, such as battery energy storage systems and flexible EV charging, to effectively balance supply and demand. By forecasting the available renewable energy generation and the expected EV charging patterns, the Hybrid DERMS optimizes the utilization of these resources to ensure efficient grid operation. It coordinates the charging of EVs with the availability of renewable energy, maximizing the use of clean and sustainable power sources and reducing reliance on conventional generation. Additionally, the Hybrid DERMS considers technical constraints and grid stability criteria when optimizing the dispatch of resources. It ensures that voltage levels, line flows, and other critical parameters are within acceptable limits to maintain the integrity and reliability of the distribution grid. By dynamically adjusting the operation of resources and coordinating their interactions, the system minimizes the risk of voltage fluctuations, overloads, power losses, and other grid disturbances and undesired consequences of unmanaged charging. The optimization and control capabilities of the Hybrid DERMS enable DSOs to make informed decisions and take proactive actions to optimize grid operation. By leveraging real-time data, advanced algorithms, and coordination mechanisms, the system facilitates the integration of high amounts of EVs and their charging infrastructure into the distribution grid. It optimizes resource utilization, reduces congestion, and ensures the efficient and reliable operation of the grid in the presence of evolving EV charging patterns and intermittent renewable DERs.

In summary, the Grid Optimization and Control functionality of the Hybrid DERMS plays a crucial role in maintaining the stability and efficiency of the distribution grid. By considering the dynamic behavior of EV charging and intermittent renewable DERs, the system optimizes the dispatch of resources, balances supply and demand, and ensures grid stability while adhering to technical constraints. This optimization enables DSOs to effectively manage the integration of high amounts of EVs and their charging infrastructure, facilitating a smooth transition to a sustainable and electrified transportation system.

4.2.4. Communication with EV charging infrastructure

Communication with EV Charging Infrastructure is a crucial aspect of the Hybrid DERMS, as it establishes direct channels of communication with EV charging infrastructure, especially front-of-the-meter charging stations connected to the medium-voltage grid (level 3 chargers). This communication enables the system to gather real-time data and exert control over the charging behavior of these EVs, contributing to the overall grid management and optimization efforts. To enable seamless communication, the Hybrid DERMS utilizes standardized communication protocols, such as Open Charge Point Protocol (OCPP) or the ISO/IEC 15,118 standard, which facilitate interoperability between the DERMS and the charging infrastructure. Through these communication channels, the DERMS can access various parameters and information from the charging stations, including charging profiles, charging rates, and availability status. By receiving real-time data from the EV charging infrastructure, the Hybrid DERMS gains valuable insights into the charging patterns and behaviors of EVs connected to the grid. This information includes load forecasts, charging schedules, and current charging status. With this data, the system can effectively monitor and analyze the impact of EV charging on the distribution grid, enabling proactive grid management and optimization. Moreover, the Hybrid DERMS can exert control over the charging behavior of EVs connected to the front-of-the-meter charging stations. By sending signals and instructions to the charging infrastructure, the system can adjust charging rates, implement load shifting strategies, and manage the charging load

in response to grid conditions and system requirements. This control capability allows the DERMS to balance the charging demand with the available grid capacity, mitigate congestion, and ensure the reliable operation of the distribution grid. In the case of behind-the-meter EVs, the Hybrid DERMS interfaces with DER/EV aggregators, which act as intermediaries between the DERMS and the individual EVs. These aggregators serve as a communication bridge, allowing the Hybrid DERMS to access the flexibility of all available resources, including behind-the-meter EVs, energy storage systems, and other DERs. The DERMS communicates with the aggregator to gather real-time data on the available flexibility and potential load curtailment options. Through this communication, the Hybrid DERMS can leverage the flexibility of behind-the-meter EVs and other DERs to support grid management objectives. It can coordinate the charging and discharging behavior of distributed resources to address grid constraints, balance supply and demand, and optimize grid operation. The system can request the aggregator to adjust the charging schedules of behind-the-meter EVs or modulate their power consumption based on grid conditions and system needs. By establishing direct communication with the EV charging infrastructure and interfacing with DER/EV aggregators, the Hybrid DERMS enhances its situational awareness and control capabilities. It can gather real-time data on EV charging, actively manage the charging behavior of EVs, and access the flexibility of behind-the-meter resources. This communication enables the DERMS to make informed decisions, optimize resource utilization, and ensure the reliable integration of high amounts of EVs into the distribution grid.

In summary, the communication capabilities of the Hybrid DERMS with EV charging infrastructure are vital for effective grid management and optimization. By establishing direct communication channels with front-of-the-meter charging stations and interfacing with DER/EV aggregators for behind-the-meter EVs, the DERMS gains real-time data, exerts control over charging behavior, and accesses the flexibility of distributed resources. This communication facilitates grid monitoring, load balancing, congestion management, and optimization efforts, supporting the successful integration of high amounts of EVs into the distribution grid.

4.2.5. Flexibility management

Flexibility management plays an important role in the integration of a large number of EVs into the grid, and the Hybrid DERMS is designed to leverage the inherent flexibility of EVs as controllable resources. By effectively managing the charging and discharging of EVs, the DERMS can address grid needs and support congestion management, contributing to the overall optimization of the distribution grid. The Hybrid DERMS employs advanced algorithms and optimization techniques to manage the flexibility of EVs in a way that aligns with grid requirements and system objectives. It takes into account various factors, including charging schedules, availability of EVs, and grid conditions, to make informed decisions regarding the utilization of EV flexibility. One of the key strategies employed by the Hybrid DERMS is demand response. By dynamically adjusting the charging schedules of EVs, the DERMS can respond to changing grid conditions and demand patterns. During periods of high grid congestion or peak demand, the DERMS can reduce the charging rate or temporarily halt charging for selected EVs, thereby reducing the load on the grid and alleviating congestion. This demand response capability helps in balancing supply and demand, ensuring grid stability, and avoiding overloading of distribution infrastructure. Load shifting is another important aspect of flexibility management. The Hybrid DERMS optimizes the charging and discharging of EVs to shift the load to off-peak periods when electricity demand is lower. By incentivizing EV owners to charge their vehicles during non-peak hours, the DERMS can effectively distribute the charging load, reduce peak demand, and optimize the utilization of grid resources. Load shifting helps in maximizing the use of renewable energy generation during periods of high availability, minimizing the need for additional generation capacity, and improving the overall efficiency of the grid. Furthermore, the Hybrid DERMS considers grid conditions, such as voltage levels, line capacities, and congestion points, to effectively manage the flexibility of EVs. It ensures that EV charging and discharging operations are aligned with technical constraints and grid reliability requirements. By optimizing the utilization of EV flexibility, the DERMS minimizes the risk of grid overloads, voltage violations, and other operational issues that may arise due to the integration of high numbers of EVs.

In summary, by harnessing the inherent flexibility of EVs, the DERMS optimizes their charging and discharging behavior to address grid needs and support congestion management. Through demand response, load shifting, and consideration of grid conditions, the DERMS ensures efficient grid operation while maximizing the benefits of EV integration. The flexibility management capabilities of the Hybrid DERMS play a vital role in balancing supply and demand, reducing congestion, and maintaining the reliability and stability of the distribution grid in the presence of a large number of EVs.

4.2.6. Summary of Hybrid DERMS

To summarize this section, the approach elucidated in this paper centers on the establishment of a Hybrid DERMS as a pivotal constituent to oversee the seamless integration of substantial volumes of EVs into distribution grids. This innovative framework endows the power system with a spectrum of real-time monitoring and control capabilities, encompassing state estimation, congestion management, grid optimization, interfacing with EV charging infrastructure, and the adept management of grid flexibility. Through the continuous aggregation and analysis of data sourced from diverse channels such as SCADA, AMI, and EV chargers, the Hybrid DERMS empowers DSOs with the ability to make well-informed decisions and take proactive control measures, ultimately optimizing grid operations, ensuring grid stability, and effectually orchestrating the integration of EVs. The proposed system architecture equips the system to swiftly identify and ameliorate peak load occurrences and congestion challenges, facilitating judicious resource utilization, and bolstering the reliability and efficiency of grid operations, even in the presence of a high amount of EVs. The proposed Hybrid DERMS framework, for real-time management of high amounts of EVs, is presented in Fig. 6.

In the subsequent section, a tangible illustration of this concept is presented through a real-world case study, describing the strategic processes entailed in the planning, integration, and proficient administration of two EV charging stations within an established distribution feeder, as well as an entire substation. This comprehensive endeavor is executed by harnessing the capabilities of an industrial DERMS solution, thus offering practical insights and empirical evidence into the intricate orchestration of such a complex undertaking within the operational power systems.

5. Case studies

In this section foure real-life examples of using the proposed Hybrid DERMS concept are presented. These use cases have undergone robust testing of integration of new EV charging stations onto an existing distribution feeder, as well as of using these EV charging stations as flexibility resources in actively managing distribution grid constraints. Use cases are presented and discussed for an example of a single feeder, as well as for an entire supply substation.

For implementation, EcoStruxure DERMS has been used [24,53], a state-of-the-art Utility DERMS solution available on the market, integrated with multiple EV Aggregators. The following test cases were conducted in an industrial setting, specifically a factory environment, and have yielded highly encouraging outcomes. Consequently, based on the promising results obtained, plans are underway to implement integration framework in various academic and industrial pilot projects. The use of EcoStruxure DERMS serves as a robust example of a commercially available Utility DERMS solution, allowing authors to effectively present

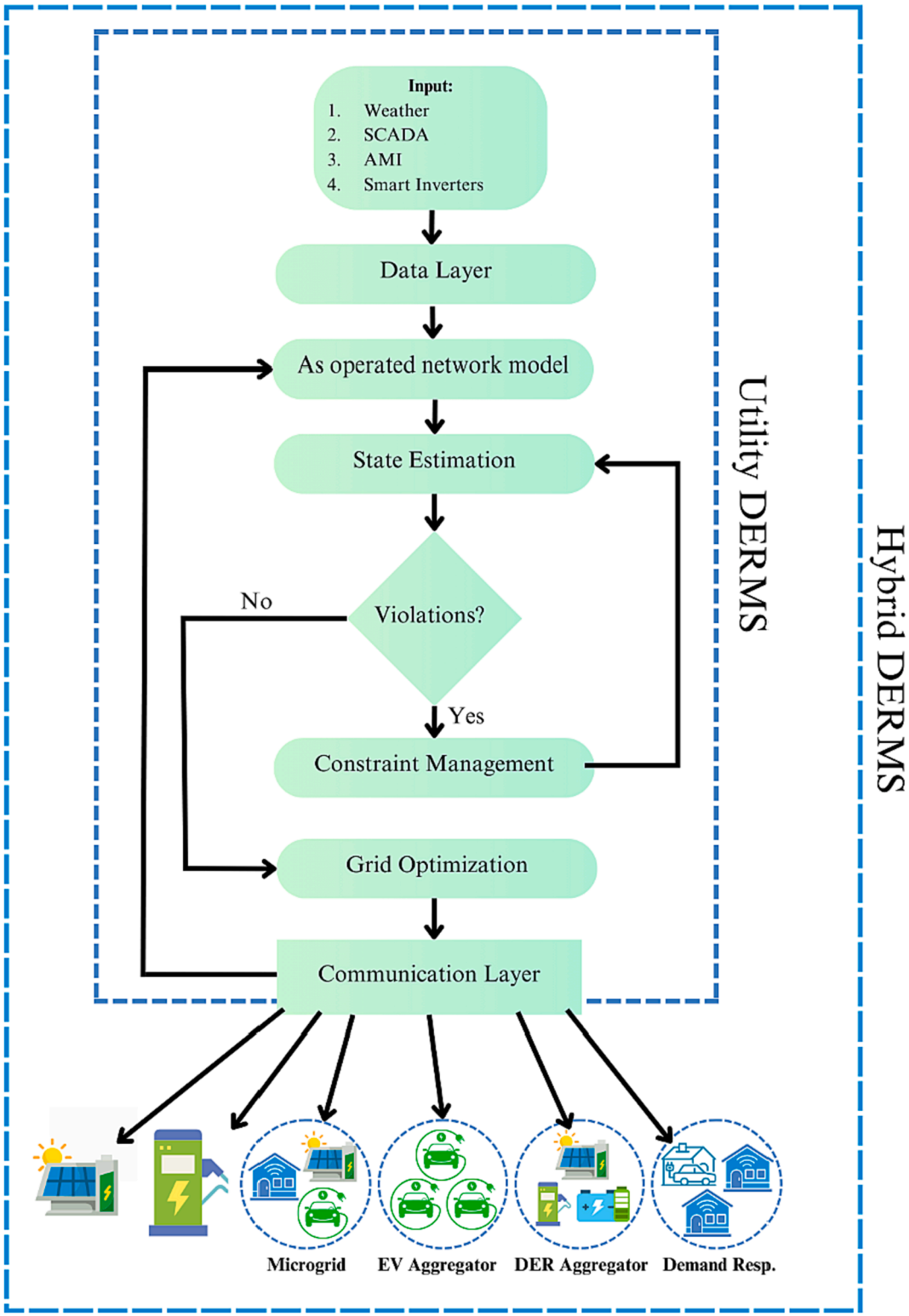


Fig. 6. Real-time management of high EV penetration through Hybrid DERMS.

their claims.

Test cases were first performed on a real-life feeder, presented in Fig. 7. The feeder supplies 185 distribution substations with 1300 end customers in total. DER (roof top solar panels) penetration is about 9 % of the total load.

The base case scenario – feeder loading profile before adding any EV charging stations is presented in Fig. 8. The maximum feeder load is 5.5 MW. The capacity of the feeder is 10.8 MW, which is determined based on the rated capacities of the equipment (transformers, lines, and cables). Moreover, other technical constraints, such as voltages, supply transformer's loading, and phase imbalance indices, are also within the required boundaries.



Fig. 7. Test feeder in this implementation.

5.1. Planning the integration of new EV charging stations

A scenario is considered where there are two EV charging stations, with a capacity of 5 MW each, that are planned to be connected to the test feeder. To analyze the impact of these connections, a DERMS planning tool is utilized. The results, illustrated in Fig. 9, show that the addition of these two EV charging stations at selected locations on the feeder would lead to a section capacity issue. The maximum loading of these sections would increase up to 125.27 % after the connection of the two EV charging stations. It is important to note that accompanying results within the same illustration manifest that no transgressions of other technical constraints—specifically pertaining to voltages, transformer loading, and the phase imbalance index—have been identified. Consequently, ensuing analyses will be exclusively focused on rectifying the identified overload issues inherent to the feeder. Nevertheless, had there been violations of any additional technical constraints, analogous analyses aimed at their resolution would have been promptly accessible through a Hybrid DERMS solution, as will be discussed in the subsequent subsections.

Fig. 10 illustrates the feeder's loading profile after the addition of two EV charging stations. The calculation yields the maximum loading of the feeder is projected to increase up to 14.7 MW during certain periods, exceeding the maximum capacity of the feeder during these critical periods. Next addressed is how identified capacity issues could be solved through the EcoStruxure DERMS.

By utilizing the EcoStruxure DERMS planning tool, a network planner has access to different strategies to explore for mitigation of the identified capacity issue. One potential strategy is network reinforcement, which involves replacing overloaded lines with stronger ones, which was the first step performed in this study. The results of the proposed replacement strategy are illustrated in Table 2.

From a technical perspective, the network reinforcement strategy provides a sustainable and permanent solution to address the capacity issue. However, this strategy faces significant challenges, including

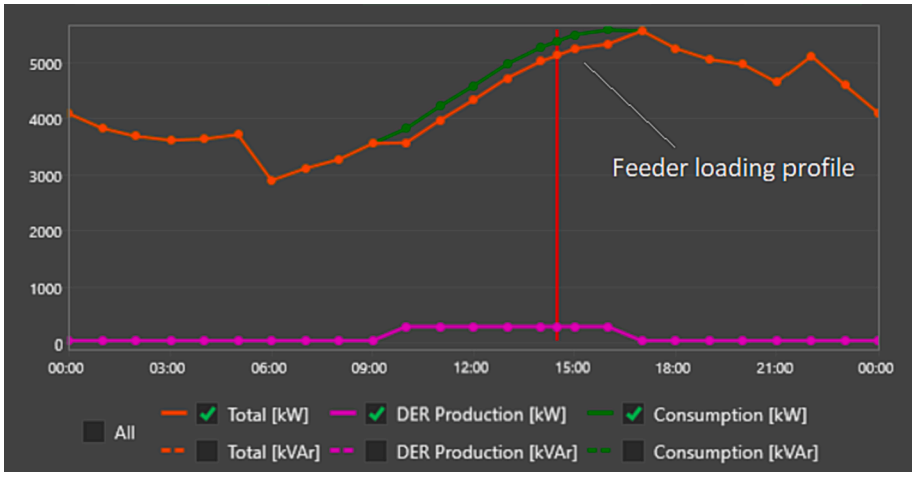


Fig. 8. The base case scenario – feeder loading.

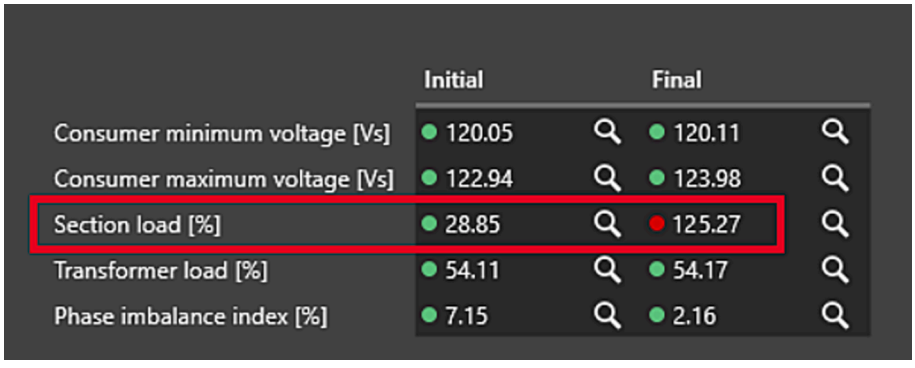


Fig. 9. Feasibility analysis of adding two EV charging stations at the test feeder.

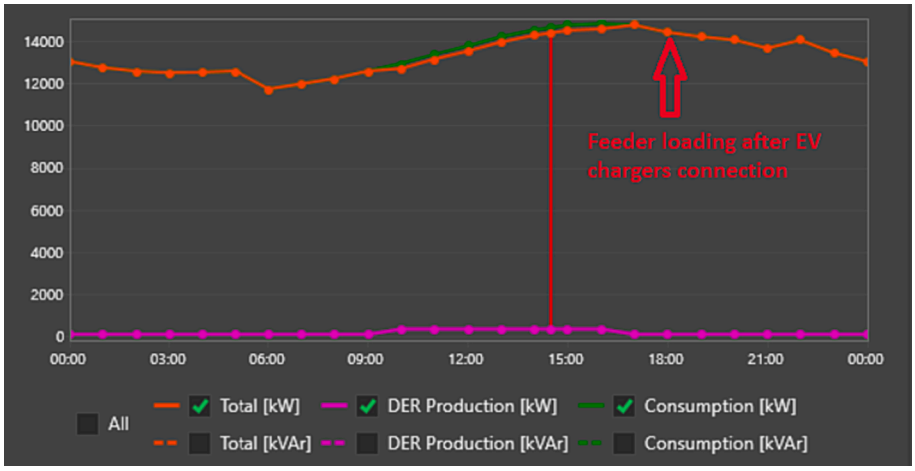


Fig. 10. The feeder profile after adding two EV charging stations.

Table 2
Network reinforcement to address violation issues upon proposal to connect two EV charging Stations.

	Initial case				**Final case**			
	Loading [%]	Loading [A]	**Rated current [A]**	Limit violation	Loading [%]	Loading [A]	**Rated current [A]**	Limit violation
Critical section	125.27	627.4	**500**	High	97.96	627.43	**640**	No

substantial capital investment required to accommodate EV connections, as discussed in Section III. Such a large investment could result in increased energy prices and prolonged customer connection times, as network reinforcement projects typically span 3 to 5 years. This means that integration of these charging stations, and consequently a substantial share of new EVs, would be forced to wait at least 3 to 5 years before they would be allowed to connect to the electrical system.

To enable faster connections while the network is being reinforced, a DSO could consider implementing a flexible interconnection approach as a non-wire alternative to network reinforcement. This approach introduces a novel paradigm for EV connections, rendering them feasible even within pre-existing network constraints. It operates within the confines of import limitations from the grid to avert potential feeder capacity challenges during periods of heightened load demand. The adaptable interconnection framework empowers consumers to maximize their energy imports under standard circumstances. However, during periods of elevated network loading, a directive from the DSO would prompt these consumers to curtail their energy imports to a predefined value. These EV charging stations, operating under flexible interconnection agreements, subsequently assume the role of agile assets for proactive network management.

In the following subsection, the concept of active network management is discussed through the lens of a practical case study, wherein two EV charging stations, aggregated via EV Aggregators, serve as dynamic resources. It is imperative to note that our discussion presupposes the mutual agreement between the charging station developers and the utility to establish flexible contracts, with the utility harnessing this flexibility through EV Aggregators.

5.2. Active network management with EVs for congestion management

First, the system is configured as follows. Initially, two EV Aggregators are added to the system. These aggregators are responsible for managing all EVs that charge through EV Charging Station 1 and EV Charging Station 2, respectively. Then, the aggregators are provided with permissions to manipulate the schedules of individual EVs within their contractual limits (through separate contracts with EV owners), in critical periods of a day (high loading), transforming the EVs into flexible resources that can be used for DSO's needs. Finally, both aggregators are integrated with a EcoStruxure DERMS (which is a Utility DERMS solution), into a Hybrid DERMS framework, proposed in this paper.

Second, network planning module within the EcoStruxure DERMS is used, but in this case, NWA is chosen instead of network reinforcement. The system now recognizes EV Aggregators as flexibility resources, and in the identified critical period when the grid is overloaded, DERMS suggests partially shifting EV charging to a less critical period and in that way avoiding overloading the grid elements. This is possible only because there are flexibility contracts in place, allowing DERMS to use EVs and their chargers for the grid needs. Now a distribution utility is ready to approve integration of new charging infrastructure to be immediately built (without a need to defer it until the grid is reinforced), but provisionally to signing a flexible contract between a DSO and EV Aggregators that allow using these chargers and shifting their consumption during identified critical periods. Thus, these two charging stations are added to a network model, and they become visible as new additions on our test feeder.

Next, an operational mode of the EcoStruxure DERMS is entered (a so-called Real-Time Module), from which a grid operator can control and manage electrical distribution grid in real-time. When the critical operation period arrives, already identified in the planning module, DERMS' Demand Flexibility application can be entered, which provides forecasted loading, as well as loading limitations. Additionally, this application shows available flexibility resources and provides an option to schedule the usage of their flexibility in order to proactively solve identified violations in the future. In Fig. 11, through a Demand Flexibility application, the forecasted loading for a test feeder is represented with the yellow line. The blue line tiles indicate the increase in loading when the EV charging stations operate at their maximum capacity, while the green tiles indicate the available decrease in loading when the EV charging stations operate at their minimum capacity (zero output). The red line represents the allowed feeder loading limit, which is 10.8 MW based on the current rating of the equipment. It can be observed that during the period between 18 h and 19 h (6 pm and 7 pm), the forecasted feeder loading exceeds the allowed feeder limits. In this period, there is an available capacity to decrease the load, which amounts to 5 MW from

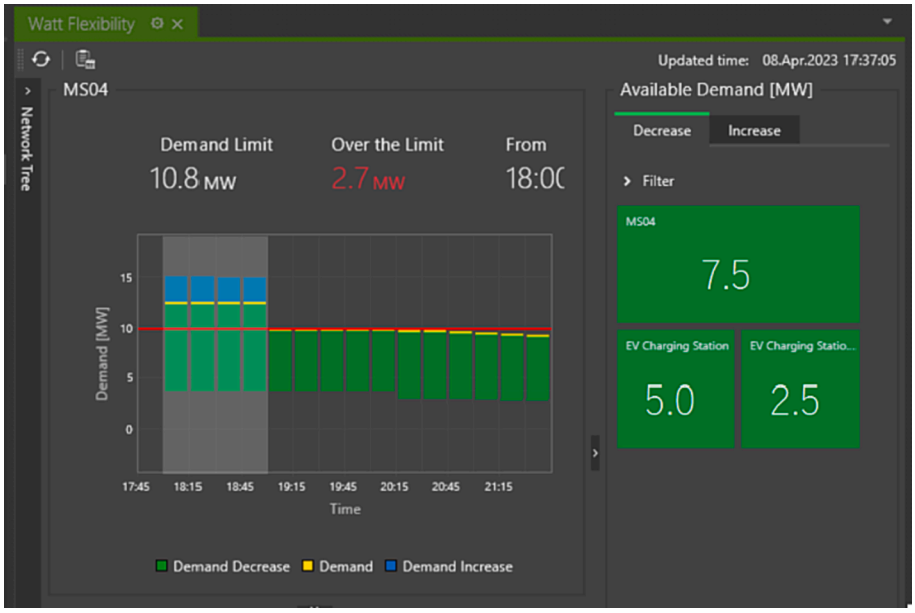


Fig. 11. Watt Flexibility in DERMS, On a Feeder Level.

the first EV charging station and 2.5 MW from the second one, totaling 7.5 MW. This available capacity is determined based on the flexible contracts between a DSO and EV Aggregators, signed in the first place, to allow integration of EVs before implementing required grid reinforcements.

To avoid a predicted overload, the Demand Flexibility within the EcoStruxure DERMS suggests to decrease charging power of the first EV charging station from 5 MW to 2.3 MW and the second one from 2.5 MW to 1.4 MW. This is further communicated to EV Aggregators, who use flexibility contracts signed with individual customers, to implement these new charging schedules. The initial and the final results (before and after using the flexibility of the EV aggregators) are illustrated in Table 3. Note that DERMS internally computes the entire state of the grid under all conditions, considering all the technical constraints, including voltages, substation loading, phase imbalance, etc., and suggests new charging schedules that will solve the detected violations while ensuring that new schedules do not cause violations of some other constraints.

By comparing the results from Table 2 and Table 3, it can be concluded that both approaches solve the technical constraint violation, but in a vastly different way. The initial methodology, as illustrated in Table 2, proposes an enhancement of the grid integrity by substituting the pivotal section(s) with novel equipment. Conversely, the subsequent strategy, outlined in Table 3, addresses the predicament by leveraging the adaptability inherent to the aggregation of EVs through the concept of flexible contracts, all the while maintaining the pre-existing equipment.

To summarize the presented case study, it involves integration of two EV charging stations, each of a 5 MW capacity, into a real-life test feeder. Using a DERMS planning tool, it is determined that connecting these stations would result in a capacity issue, causing the maximum loading of a certain sections to increase up to 125.27 %. The feeder's loading profile after the addition of the charging stations shows that the maximum loading would exceed the feeder's capacity during specific periods, reaching 14.7 MW. To address this capacity issue, network reinforcement is proposed as a technical solution, but it presents challenges such as substantial capital investment, increased energy prices, and prolonged customer connection times. Alternatively, a flexible interconnection approach is suggested as a non-wire alternative, allowing EV connections even in constrained networks with import limitations during high network loading. This approach utilizes EVs and their chargers as flexible resources for active network management. Through the integration of EV Aggregators with a Utility DERMS, into a Hybrid DERMS solution, the system can manipulate the schedules of individual EVs to avoid overloading the grid during critical periods. This enables the immediate integration of new charging infrastructure with the provision of flexible contracts between the DSO and EV Aggregators (as well as aggregators and EV owners). Furthermore, the DERMS's real-time and lookahead modules, including the presented Demand Flexibility application, allow grid operators to control and manage distribution grid, forecast loading, and schedule the usage of flexibility resources to proactively address violations.

In the following subsection, the concept of active network management is discussed through the lens of managing customer voltages, wherein two EV charging stations, aggregated via EV Aggregators, serve as dynamic resources capable to maintain charging level below requested value. The same assumption that the mutual agreement between the charging station developers and the utility to establish flexible contracts, with the utility harnessing this flexibility through EV Aggregators is made here, as in the previous cases.

5.3. *Active network management with EVs to improve customer voltages*

Two EV Aggregators are considered as in the case V.B. These aggregators are responsible for managing all EVs that charge through EV Charging Station 1 and EV Charging Station 2, respectively. Then, the aggregators are provided with charging limits of individual EVs, in critical periods of a day (low voltages), transforming the EVs into

Table 3
NWA to address violation issues upon proposal to connect two EV charging stations.

	Initial case				**Final case**			
	Loading [%]	**Loading [A]**	Rated current [A]	Limit violation	Loading [%]	**Loading [A]**	Rated current [A]	Limit violation
Critical section	125.27	**627.4**	500	High	89.70	**448.5**	500	No

flexible resources that can be used for DSO's needs. Finally, both aggregators are integrated with a EcoStruxure DERMS (which is a Utility DERMS solution), into a proposed Hybrid DERMS framework.

EV Charging Schedule (portion of it) is presented in Fig. 12. These schedules are communicated by EV aggregators to the EcoStruxure DERMS.

Accounting for the EV Charing Schedules, EcoStruxure DERMS identifies customer voltage issues at the feeder as illustrated in Fig. 13. It is assumed that the allowed customer voltage range is between 95 % and 110 %.

Triggering on customer voltage issues, EcoStrxure DERMS identifies EV Charging Stations as flexible resouces at the affected feeder and recalcuates charging limit for them. Calculated charging limits are now 2.5 MW and 2 MW, respectively. Upon verification that new limits will not create any new issue in the system, EcoStruxure DERMS subsequently conveys the revised limits to both aggregators. Upon applying new limits, customer voltage profile is updated as illustrated in Fig. 14.

Finally, to showcase the applicability of the proposed Hybrid DERMS concept in managing EVs on a larger scale, in the next section we present a use case that includes managing the peak load on a substation level, by shifting the EV charging time through Hybrid DERMS.

5.4. Managing peak loads caused by EVs on a supply substation level

In Fig. 15, we depict the Demand Flexibility application inherent to the EcoStruxure DERMS, but this representation is contextualized at the supply substation level, subsequent to the integration of the previously delineated two large EV charging stations.

As illustrated in Fig. 15, when one scrutinizes the current charging schedules of the EV charging stations—communicated by the EV aggregators to the EcoStruxure DERMS—it becomes evident that the peak load is anticipated between 18:00 (6 pm) and 19:00 (7 pm). In contrast, the nadir of demand is discerned between 16:15 (6.15 pm) and 17:00 (5 pm). Given that the flexibility inherent to EV Charging Station 2 permits the reallocation of a portion of its charging from the observed peak load period to intervals of lower demand, the EcoStruxure DERMS recommends transposing the EV charging to off-peak durations, as illustrated in Fig. 16.

Prior to dispatching the updated schedule for implementation by the EV Aggregator, the EcoStruxure DERMS executes a rigorous computational assessment of all pertinent technical constraints, similar to the previously presented use cases for one feeder, in alignment with the proposed schedule. Upon verification that no constraints are infringed, the DERMS subsequently conveys the revised schedule to the aggregator.

The initial and the final state, regarding the peak load are presented in Table 4.

In summary of the delineated case study, it encompasses the integration of EV charging stations within an operational test supply substation equipped with a transformer boasting a rated power of 25MVA. Leveraging a Hybrid DERMS solution, the supply transformer's loading is ascertained and continuously monitored for a prospective duration. In a bid to equilibrate transformer loading and circumvent peak demands, the system adeptly adjusts individual EV schedules to reallocate load to off-peak intervals or moments of maximal renewable production. This stratagem facilitates the prompt assimilation of new charging infrastructures, underpinned by malleable contracts between the DSO and EV Aggregators, as well as between aggregators and the individual EV owners.

6. Conclusions

The integration of EVs into electrical distribution grids presents both opportunities and challenges. While EV technology has rapidly evolved, overcoming barriers such as charging time, battery capacity, and cost, the increased adoption of EVs raises concerns about the safety and reliability of electrical distribution grids. Traditional passive distribution networks designed for one-way power flow are ill-equipped to handle the dynamic and stochastic nature of EV charging. Uncontrolled integration of EVs may lead to issues such as peak load rise, voltage fluctuations, feeder overloads, and power losses. However, the existing body of research indicates that with intelligent planning and optimized management, EVs can serve as valuable flexibility resources for DSOs. The benefits of proper EV charging coordination has been also highlighted, including voltage improvements, congestion management, voltage regulation, and reactive power compensation. To harness these benefits, in this paper, a new framework called Hybrid DERMS is proposed as a solution.

Our primary objective is to address the complex challenges associated with the mass integration of EVs into emerging distribution grids. DERMSs are pivotal in the planning, integration, and management of distributed energy resources, including EVs. Recognizing the evolving demands of distribution grids, this paper introduces a Hybrid DERMS framework, integrating various hierarchical levels of DER management solutions. This innovative framework is designed to effectively navigate the intricacies posed by the large-scale integration of EVs. Our approach

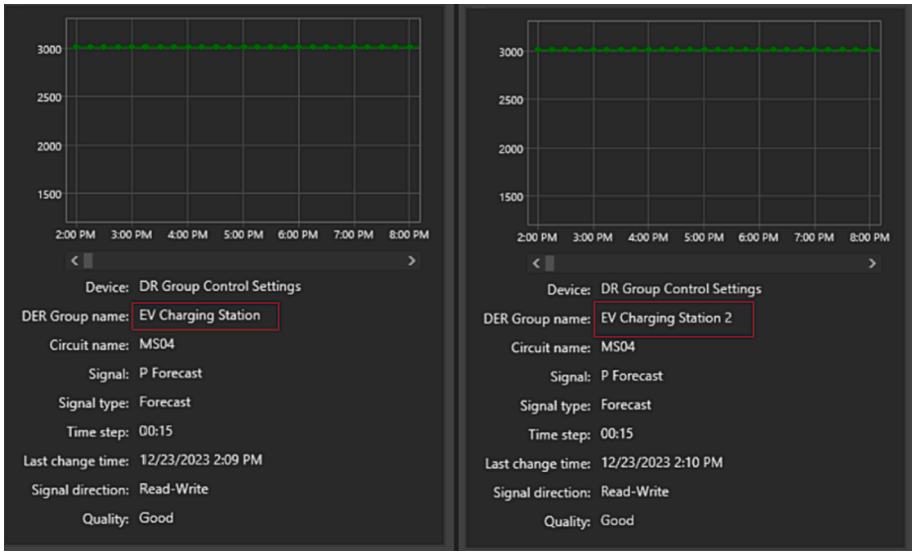


Fig. 12. Calculated voltage profile for feeder with two EV charging stations.

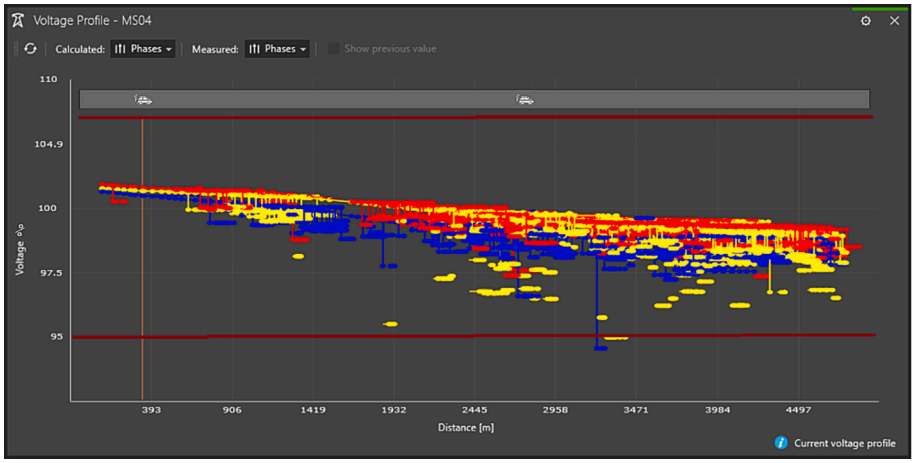


Fig. 13. Calculated voltage profile for feeder with two EV charging stations.

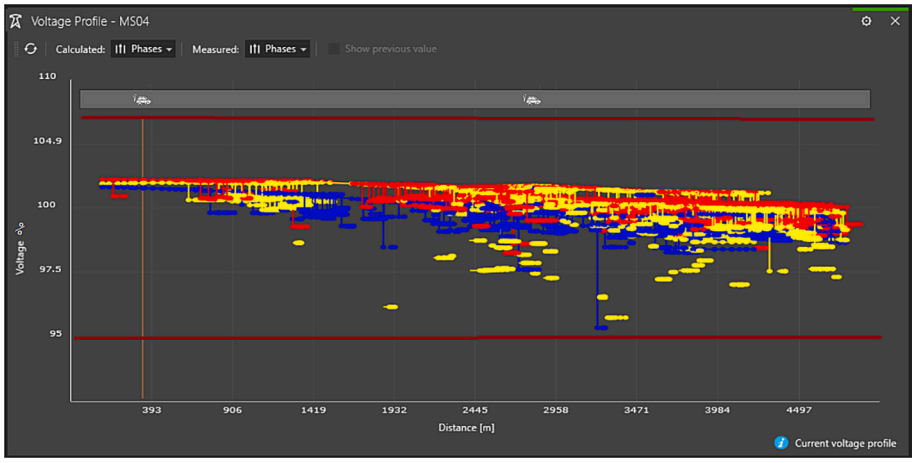


Fig. 14. Calculated voltage profile for feeder with two EV charging stations upon voltage regulation through Hybrid DERMS.

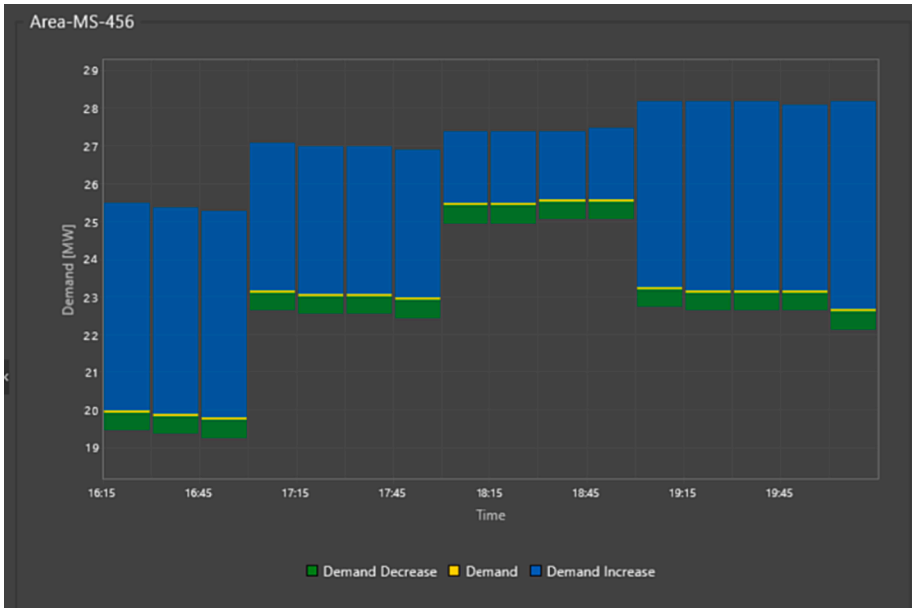


Fig. 15. Watt Flexibility in DERMS, On a Substation Level.

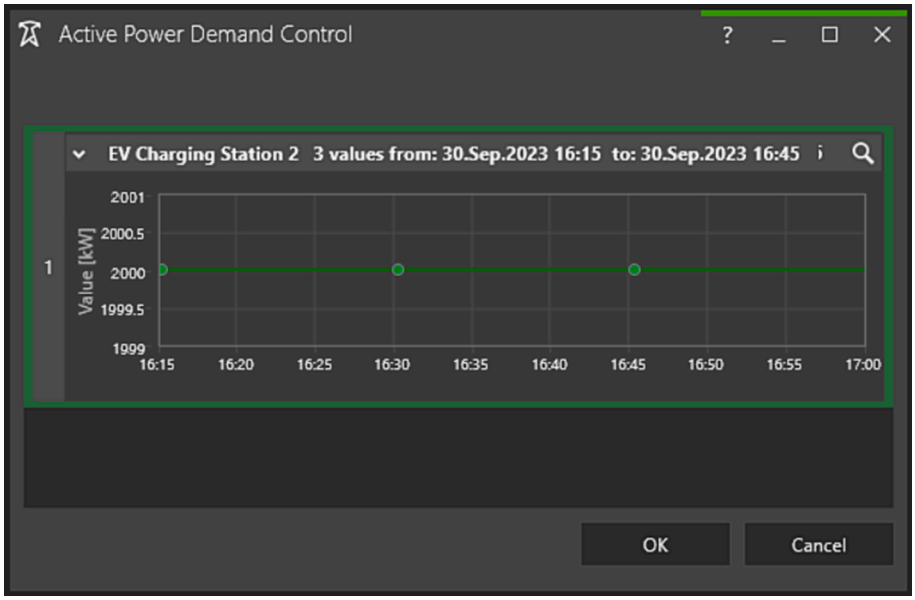


Fig. 16. New Schedule for the EV Charging Station 2.

Table 4
Peak Load Management Through Shifting EV Charging Schedule.

	Initial case				Final case			
	Peak load [MW]	Peak load period	Percent of a maximal load	Limit violation	Peak Load [MW]	Peak load period	Percent of a maximal load	Limit violation
Supply substation	25.7	**6 pm-7 pm**	101 %	Low level violation	22.1	**4 pm-5 pm**	84 %	No

empowers DSOs to safely plan and intelligently manage distribution grids with high volumes of EVs. This is achieved while concurrently avoiding technical violations, thereby ensuring grid stability and reliability. The paper delves into the planning and operational challenges faced by DSOs and engineers in the context of mass EV integration. It presents a comprehensive analysis of various DERMS solutions, assessing their strengths and weaknesses. A key component of this study is a real-life case study, demonstrating the successful integration and real-time management of distribution feeders with significant shares of EVs using a proposed Hybrid DERMS solution. The results from this case study are pivotal. They clearly illustrate that, by leveraging the capabilities of a Hybrid DERMS, DSOs can effectively integrate and manage large quantities of EVs. This not only ensures the efficient and safe operation of distribution grids but also significantly contributes to the electrification and decarbonization of the transportation sector. In conclusion, the main contribution of this paper is the introduction and validation of the Hybrid DERMS as a comprehensive tool for DSOs, offering a strategic solution to the challenges brought forth by mass EV integration in distribution grids.

It is important to acknowledge that once a certain threshold of EVs is reached, the existing grid infrastructure may necessitate hardware upgrades to enhance its capacity and adequately support the complete electrification of transportation. Nevertheless, the implementation of a Hybrid DERMS that intelligently manages the evolving distribution grid holds significant potential for DSOs to address the transitional phase towards a future where EVs comprise a substantial portion of the vehicle fleet. This approach enables DSOs to bridge the gap between the present and the inevitable future characterized by high EV penetration, without the requirement to postpone the further deployment of EVs until the grid is fully reinforced. By leveraging a Hybrid DERMS, DSOs can optimize grid operations and effectively balance the increasing EV demand with available grid resources, ensuring security and efficient power distribution without compromising the system reliability.

CRediT authorship contribution statement

Luka Strezoski: Conceptualization, Formal analysis, Investigation, Methodology, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Izabela Stefani:** Data curation, Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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